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# AIM:
# Facial Recognition Levels of Computer Vision
# (Image Acquisition -> Face Detection -> Decision)
# =====

import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload Image
print("Upload a face image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read Image
img = cv2.imread(filename)
rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Load Haar Cascade Face Detector
face_detector = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_frontalface_default.xml"
)

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Detect Faces
faces = face_detector.detectMultiScale(
    gray,
    scaleFactor=1.2,
    minNeighbors=5
)

# Draw Face Box
for (x, y, w, h) in faces:
    cv2.rectangle(rgb, (x, y), (x+w, y+h), (0,255,0), 2)
    cv2.putText(rgb,
        "Face Detected",
        (x, y-10),
        cv2.FONT_HERSHEY_SIMPLEX,
        0.8,
        (255,0,0),
        2)

# Display Image
plt.figure(figsize=(7,7))
plt.imshow(rgb)
plt.title("Facial Recognition")
plt.axis("off")
plt.show()

# Decision Making
print("\n===== COMPUTER VISION REPORT =====")
print("Low-Level Vision : Image Captured")
print("Mid-Level Vision : Face Detected")

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if len(faces) > 0:
    print("High-Level Vision : Face Present")
    print("Decision          : Authorized User (Demo)")
else:
    print("High-Level Vision : No Face Found")
    print("Decision          : Access Denied")

print("=====")
```

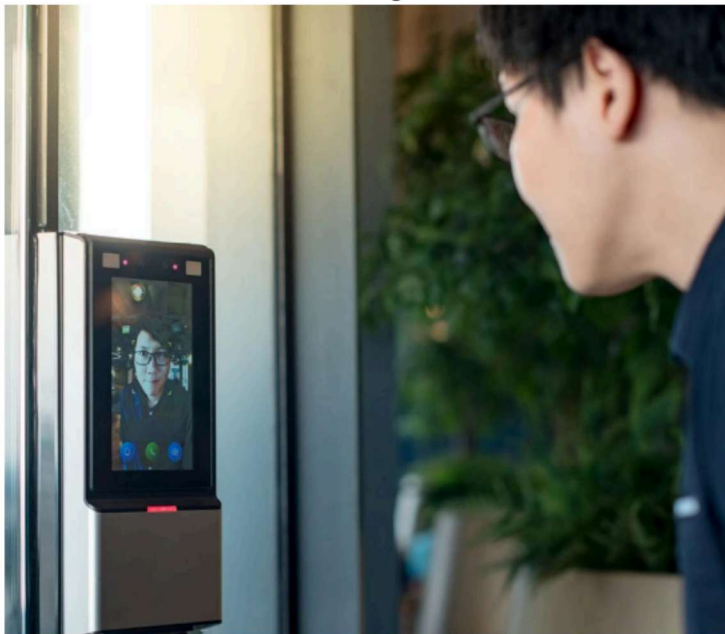
Upload a face image

Q9.jpg

Q9.jpg(image/jpeg) - 44187 bytes, last modified: 7/15/2026 - 100% done

Saving Q9.jpg to Q9.jpg

Facial Recognition



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===== COMPUTER VISION REPORT =====
Low-Level Vision : Image Captured
Mid-Level Vision : Face Detected
High-Level Vision : No Face Found
Decision         : Access Denied
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